

Trigonometry

Week 1D

Overview

Trigonometry

Key Ideas - Graphs of Trigonometric Functions

Key Idea – Graphs of Trigonometric Functions using Radians

Introduction:

<https://www.desmos.com/calculator/hptqxr1k0>

Discussion Questions:

- (a) Use Desmos (<https://www.desmos.com/calculator>) to graph the functions $y = \sin x$ and $y = \cos x$.

In the graph settings, ensure that the angle is set to radian and set the x -axis step to π .

What is the range of the two graphs? *range $\rightarrow [-1, 1]$*

After what interval on the x -axis does each graph repeat itself? *0 to 2π*

What is the horizontal shift between the two graphs? *$\frac{\pi}{2}$*

- (b) Use Desmos to graph $y = \tan x$.

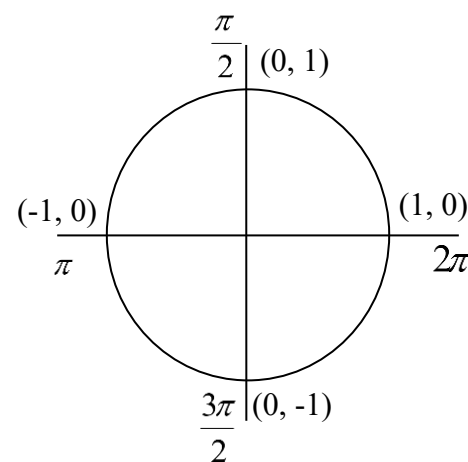
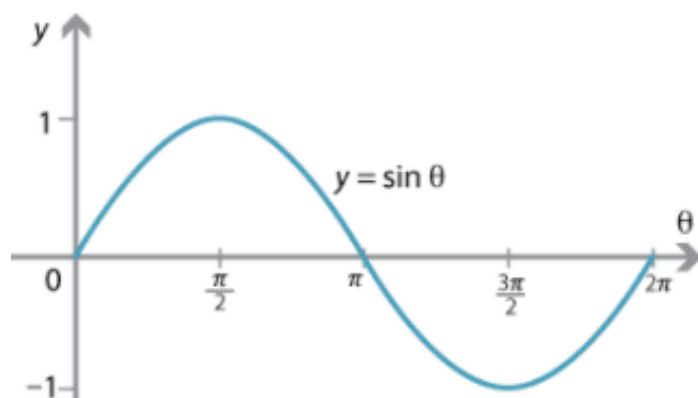
What is the range of the graph? *range $\rightarrow (-\infty, \infty)$*

After what interval on the x -axis does the graph repeat itself? *$-\pi$ to π*

The graph $y = \sin x$

To obtain the graph of $y = \sin x$, the **y-coordinates** are plotted as the angle varies on the unit circle from $0 \leq x \leq 2\pi$.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
y	0	1	0	-1	0

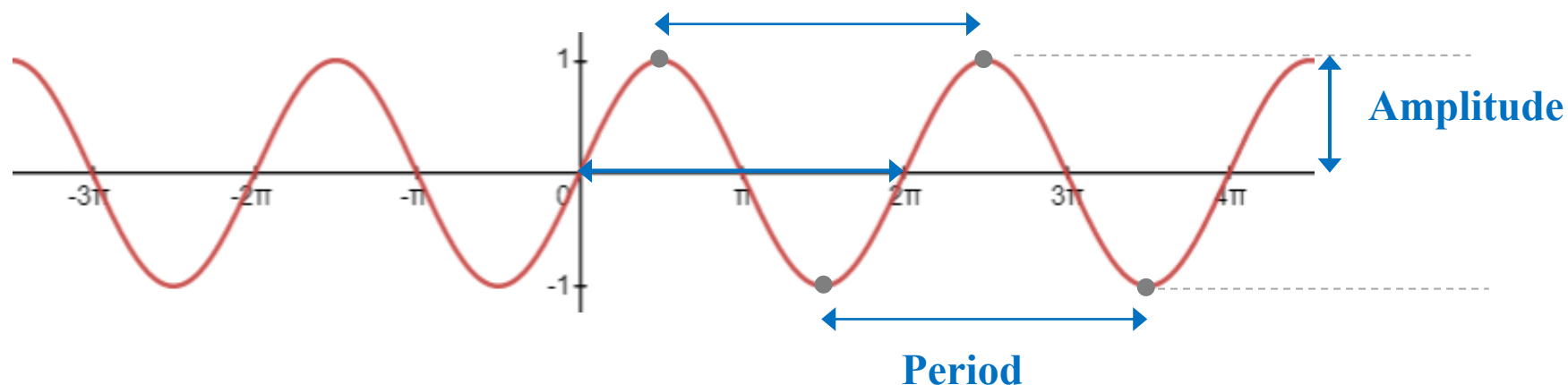


The graph for $0 \leq x \leq 2\pi$ forms a cycle which is then repeated.

Since the graph of the sine function replicates itself for every 2π along the x -axis:

$$\sin(x + 2k\pi) = \sin x, \text{ for } k = 0, \pm 1, \pm 2, \pm 3, \dots, \dots,$$

Graph of $y = \sin x$, with the x -axis labelled in pi radians



The **period** p of a trigonometric function is the distance between two successive crests or troughs (or from any point to the next matching point) on the curve.

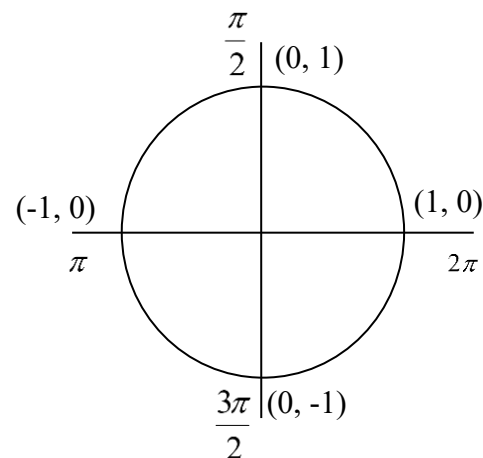
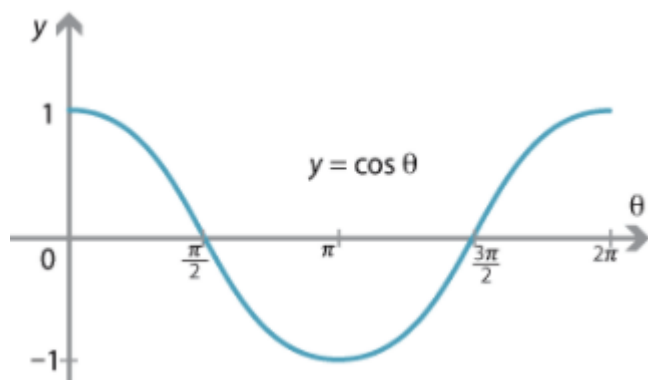
The amplitude of a trigonometric function is half the difference between the maximum and minimum values.

The sin function is **continuous** and **periodic** with a period of 2π and an **amplitude** of 1.

The Graph of $y = \cos x$

To obtain the graph of $y = \cos x$, the **y-coordinates** are plotted as the angle varies on the unit circle from $0 \leq x \leq 2\pi$.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \cos x$	1	0	-1	0	1

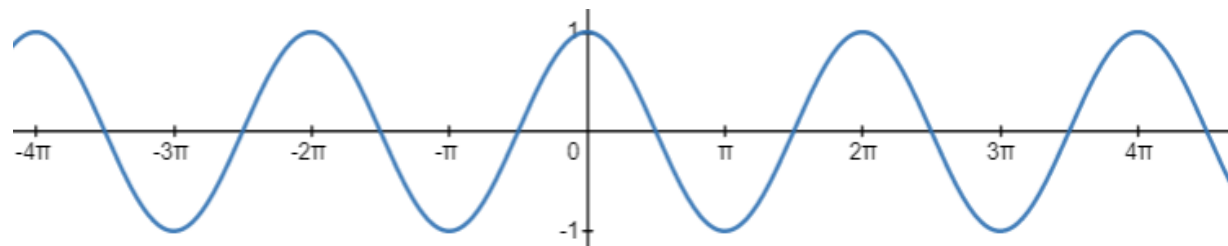


The graph for $0 \leq x \leq 2\pi$ forms a cycle which is then repeated.

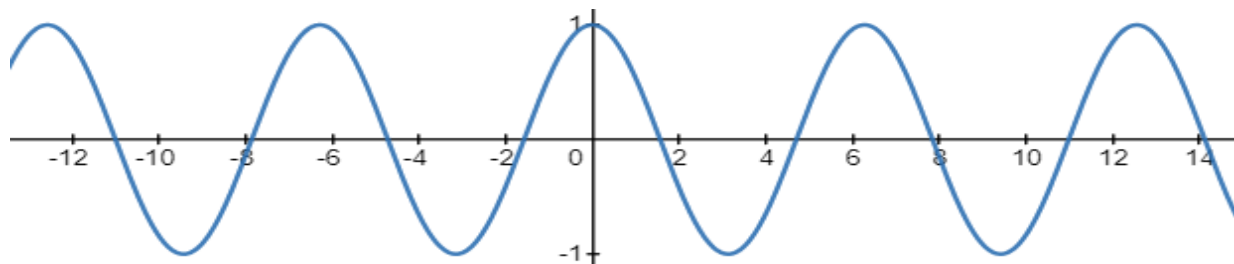
Since the graph of the cosine function replicates itself every 2π units along the x -axis:

$$\cos(x + 2k\pi) = \cos x, \text{ for } k = 0, \pm 1, \pm 2, \pm 3, \dots.$$

Graph of $y = \cos x$, with the x -axis labelled in pi radians



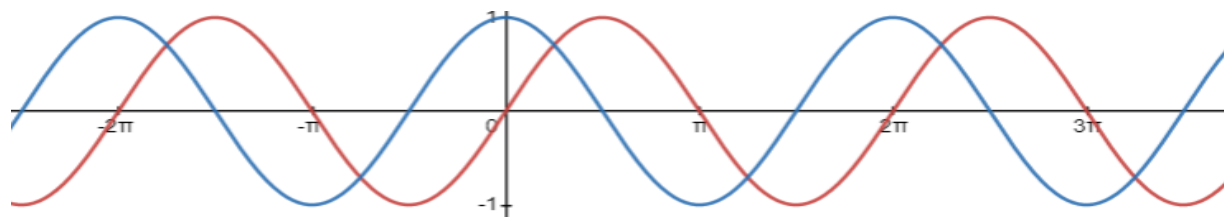
Graph of $y = \cos x$, with the x -axis labelled in radians



The cosine function is **continuous** and **periodic** with a period of 2π .

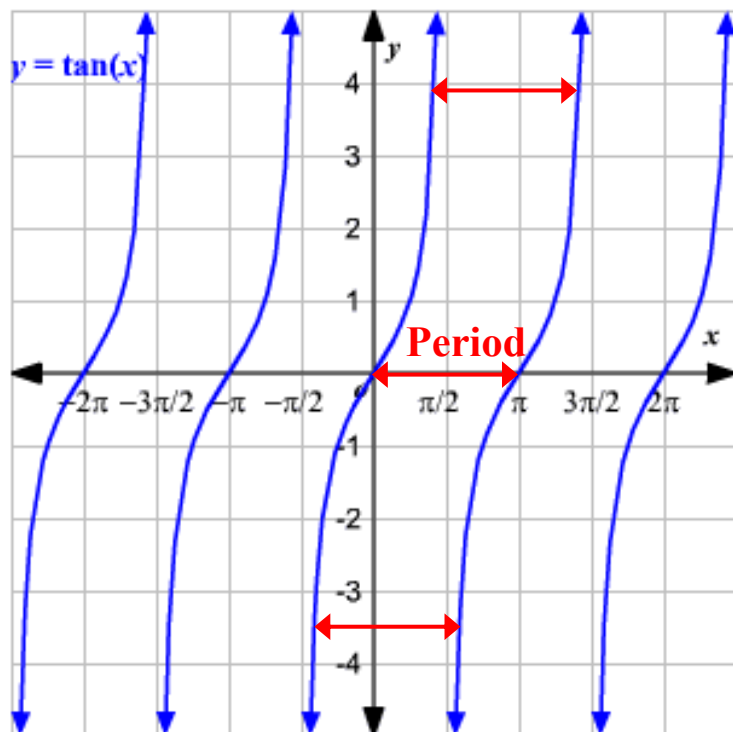
The **amplitude** of the cosine function is 1.

The graph of the cosine function is the same shape as the sine function, just shifted along the x -axis.



The Graph of $y = \tan x$

The graph of the tan function is discontinuous at $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$



The tan graph has **no amplitude** as the range is $-\infty < y < \infty$.

The tan function is periodic with a **period** π .

Since the graph replicates itself every π units along the x -axis:

$$\tan(x + k\pi) = \tan x, \text{ for } k = 0, \pm 1, \pm 2, \pm 3, \dots,$$

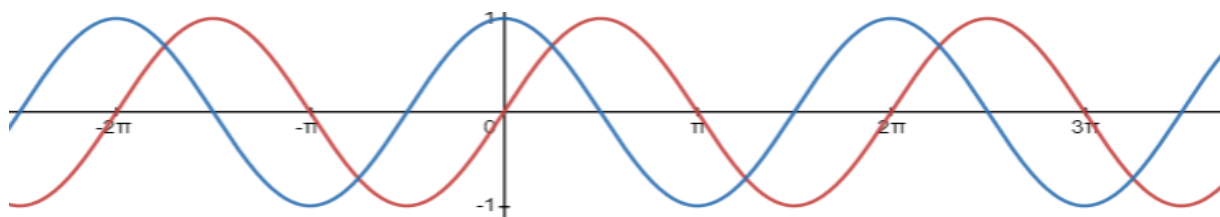
Transformation of Graphs of Trigonometric Functions

The general form of the sine function is

$$y = A \sin B(x - h) + k.$$

A sinusoid is the name given to any curve that is written in this form. The general form of the cosine function can also be written as $y = A \sin B(x - h) + k$. This is because the cosine function is identical to the sine function except for a horizontal shift.

$$\cos x = \sin\left(x + \frac{\pi}{2}\right)$$



Sinusoidal functions are also found in other fields of study apart from Mathematics, such as Science and Engineering. The sinusoidal function occurs in nature, for example in ocean waves, sound waves and graphs of average daily temperature throughout the year.

The four constants in the equation (A , B , h and k) will be analysed separately by comparing the graph of $y = \sin x$ with graphs of $y = A \sin x$, $y = \sin Bx$, $y = \sin(x - h)$ and $y = \sin x + k$.

Notice that when $A = 1$, $B = 1$, $h = 0$ and $k = 0$, the general equation simplifies to $y = \sin x$.

The graph of the function $y = \sin x$ is the parent graph for all sinusoidal graphs.

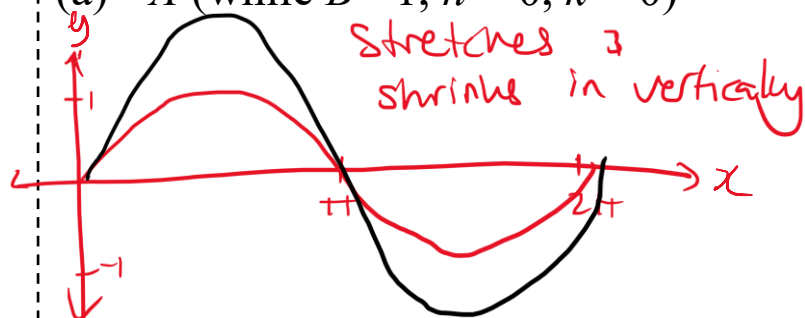
Discussion Questions: <https://www.desmos.com/calculator/g0oabzbh9w>

Use Desmos to analyse the transformations of the sine graph by changing values for A , B , h and k . Use the sliders or the animation feature to study the transformations.

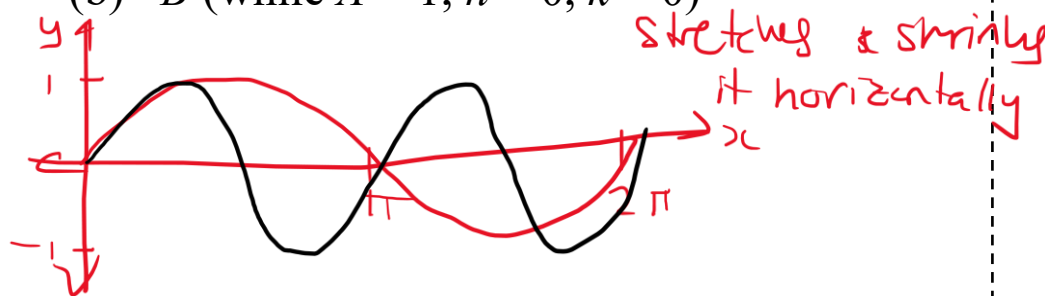
Firstly, ensure that in Desmos the constants are set to default: $A = 1$, $B = 1$, $h = 0$ and $k = 0$.

Describe the transformations resulting as each constant is changed:

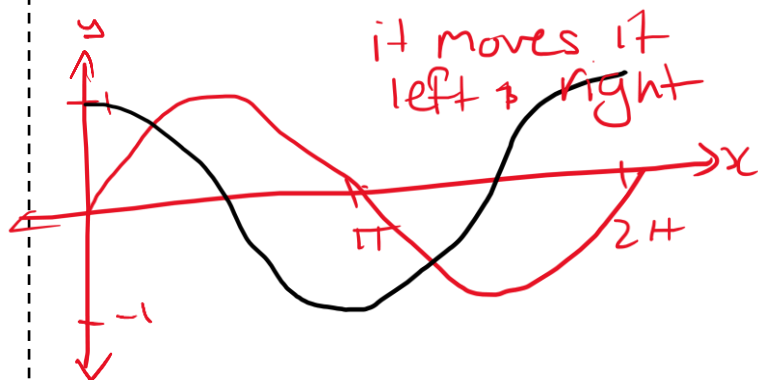
(a) A (while $B = 1$, $h = 0$, $k = 0$)



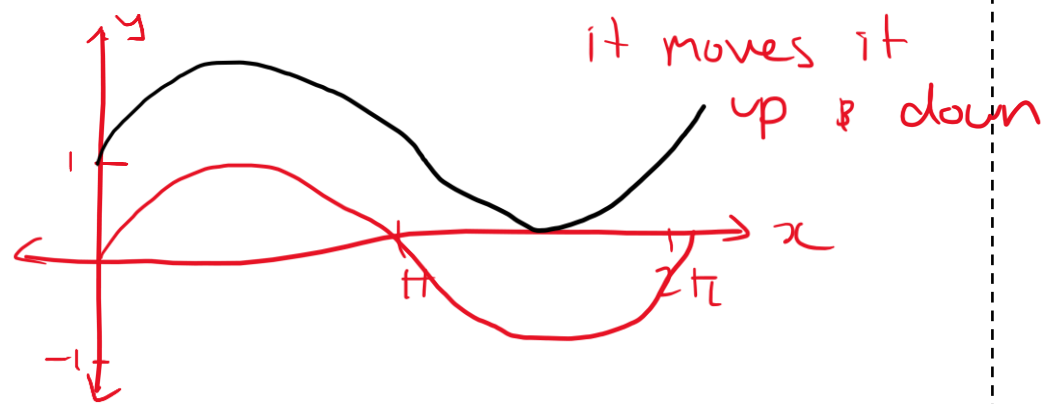
(b) B (while $A = 1$, $h = 0$, $k = 0$)



(c) h (while $A = 1$, $B = 1$, $k = 0$)



(d) k (while $A = 1$, $B = 1$, $h = 0$)



General Equation $y = A \sin B(x - h) + k$

Compared to parent graph:

A vertically stretches or shrinks the graph.

B horizontally stretches or shrinks the graph.

h horizontally shifts the graph

k vertically shifts the graph.

Each constant will be analysed separately by comparing $y = \sin x$ with

$$y = A \sin x \quad y = \sin Bx \quad y = \sin(x - h) \quad \text{and} \quad y = \sin x + k$$

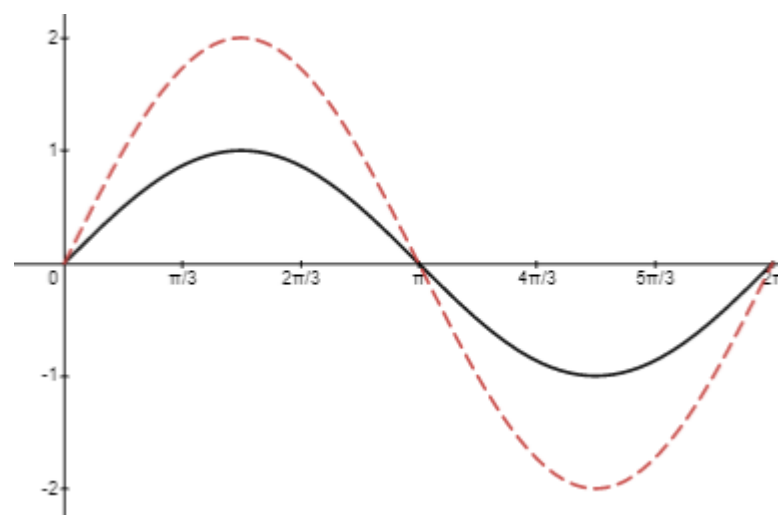
Constant A $y = A \sin x$

Consider the graphs of $y = \sin x$ and $y = 2 \sin x$

The graph of $y = 2 \sin x$ has an amplitude of 2.

The amplitude of the curve is determined by the constant A .

Altering the amplitude does **not** change the points at which the graph cuts the x -axis.

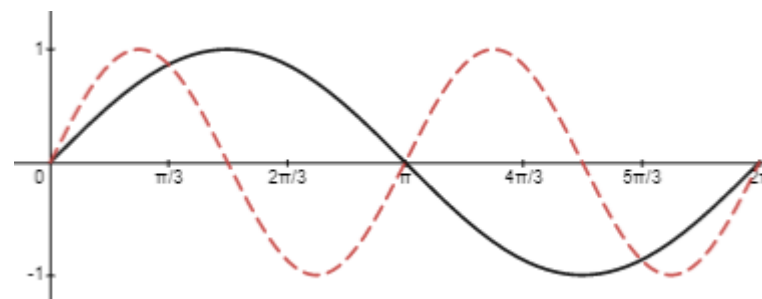


Constant B $y = \sin Bx$

Consider the graphs of $y = \sin x$ and $y = \sin 2x$

The graph of $y = \sin 2x$ has two cycles in 2π .

Therefore, the period of $y = \sin 2x$ is π which is half the period of $y = \sin x$.



The value of B determines the number of cycles in the interval $0 \leq x \leq 2\pi$.

The value of B determines the period of the graph,

given by $p = \frac{2\pi}{B}$

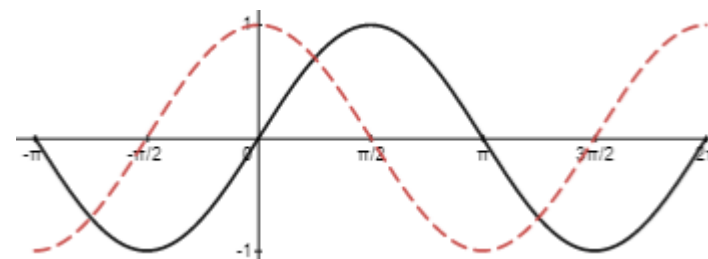
When $0 < B < 1$, the period of the function will be greater than 2π and the graph will have a horizontal stretching.

When $B > 1$, the period of the function will be less than 2π and the graph will have a horizontal shrinking.

Constant h $y = \sin(x + h)$

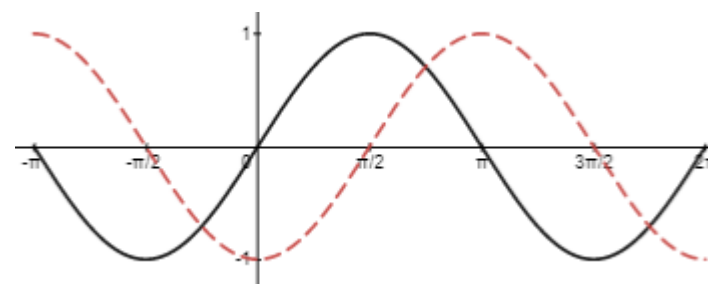
Consider the graphs of $y = \sin x$ and $y = \sin\left(x + \frac{\pi}{2}\right)$

The parent graph is shifted to the left by $\frac{\pi}{2}$ units.



Consider the graphs of $y = \sin x$ and $y = \sin\left(x - \frac{\pi}{2}\right)$

The parent graph is shifted to the right by $\frac{\pi}{2}$ units.



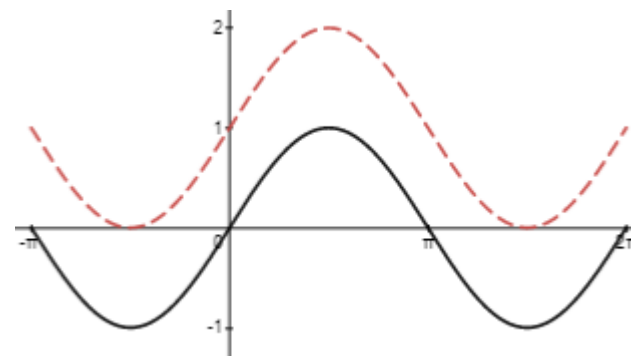
Therefore,

- horizontal shift to the left when h is positive
- horizontal shift to the right when h is negative.

Constant k $y = \sin x + k$

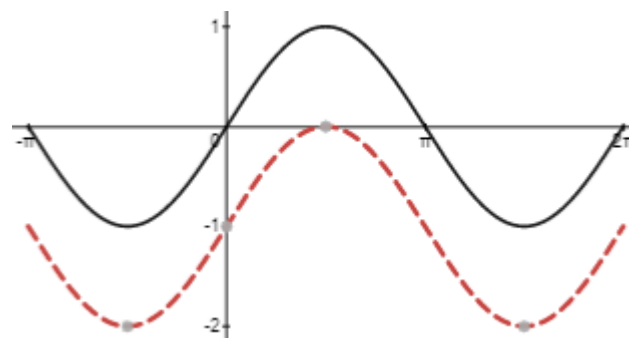
Consider the graphs of $y = \sin x$ and $y = \sin x + 1$

The parent graph is shifted up by 1 unit.



Consider the graphs of $y = \sin x$ and $y = \sin x - 1$

The parent graph is shifted down by 1 unit.

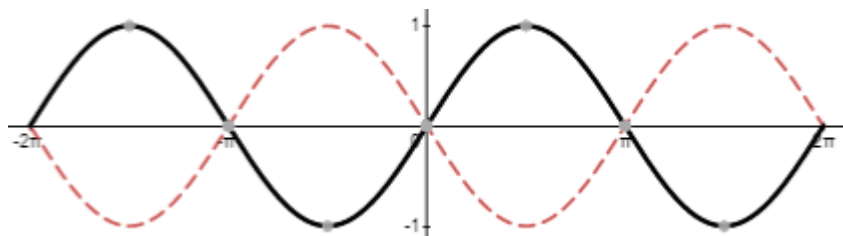


Therefore,

- vertical shift up when k is positive
- vertical shift down when k is negative.

Reflection in the x -axis

Consider the graphs of $y = \sin x$ and $y = -\sin x$.



The negative coefficient flips the graph. It is reflected in the x -axis.

In summary: General Equation $y = A \sin B(x - h) + k$

$|A|$ = amplitude

B = cycles in interval $0 \leq x \leq 2\pi$ and period $p = \frac{2\pi}{B}$

h = horizontal shift

k = vertical shift

When A is negative, the curve is reflected in the x -axis.

Exercise:

Determine the period and amplitude for each of the following functions:

(a) $y = 2\sin x$

$$|A| = |2| = 2 \rightarrow \text{amplitude} = 2$$

$$B = 1$$

$$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{1} = 2\pi$$

(c) $y = 3\sin 2x$

$$|A| = |3| = 3 \rightarrow \text{amplitude} = 3$$

$$B = 2$$

$$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{2} = \pi$$

(e) $y = 2\sin 4x + 1$

$$|A| = |2| = 2 \rightarrow \text{amplitude} = 2$$

$$B = 4$$

$$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$$

(b) $y = \cos \frac{x}{2}$

$$|A| = |1| = 1 \rightarrow \text{amplitude} = 1$$

$$B = \frac{1}{2}$$

$$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{\frac{1}{2}} = 4\pi$$

(d) $y = 4\cos 3x$

$$|A| = |4| = 4 \rightarrow \text{amplitude} = 4$$

$$B = 3$$

$$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{3}$$

(f) $y = 1 - 2\cos 4x$

$$|A| = |-2| = 2 \rightarrow \text{amplitude} = 2$$

$$B = 4$$

$$\text{period} = \frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$$

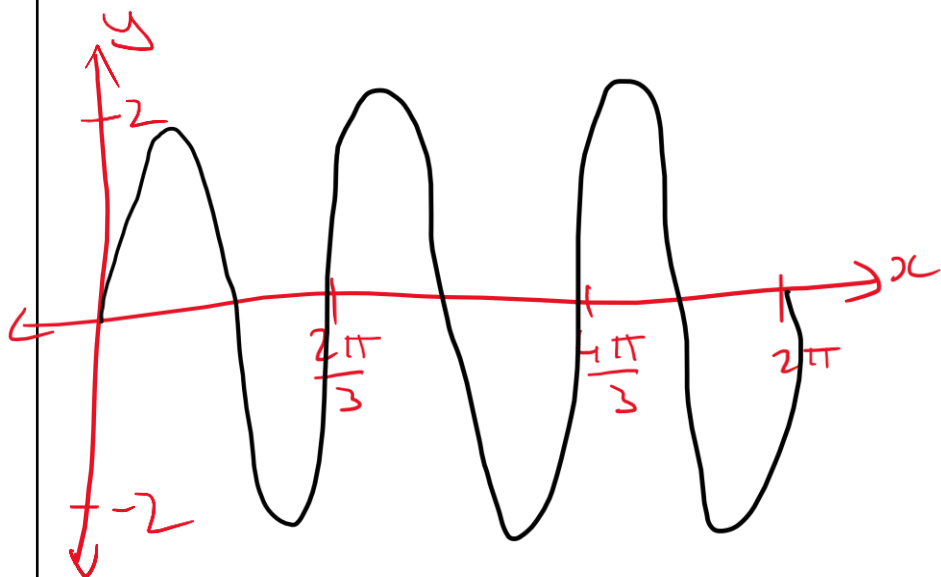
Exercise:

(a) For $0 \leq x \leq 2\pi$ sketch $y = 2\sin 3x$

• amplitude = $|A| = |2| = 2$

• period = $\frac{2\pi}{B} = \frac{2\pi}{3}$

• number of cycles in $2\pi = B = 3$

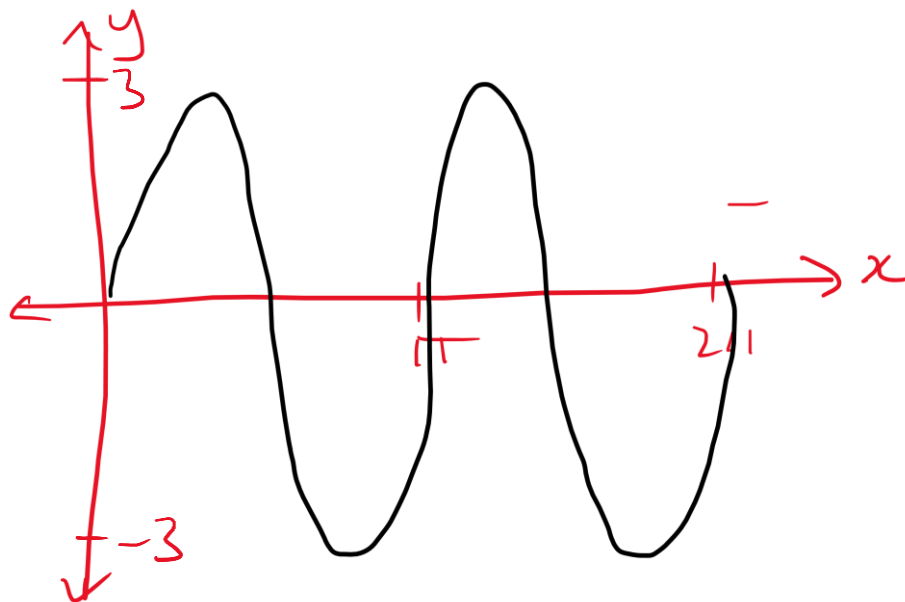


(b) For $0 \leq x \leq 2\pi$ sketch $y = 3\sin 2x$

• amplitude = $|A| = |3| = 3$

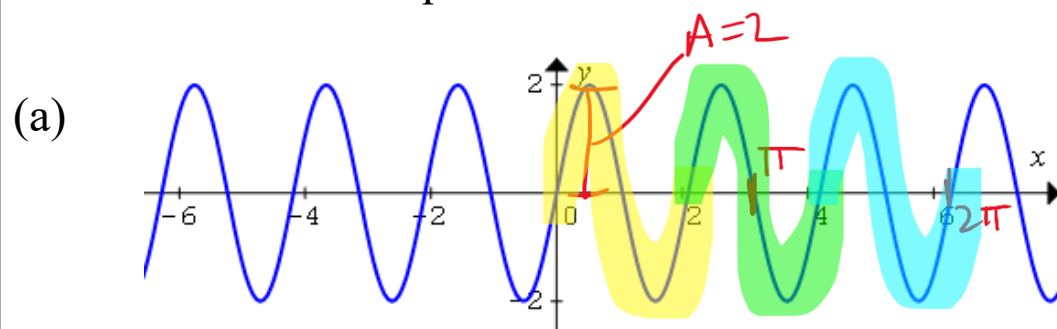
• period = $\frac{2\pi}{B} = \frac{2\pi}{2} = \pi$

• number of cycles in $2\pi = 2$



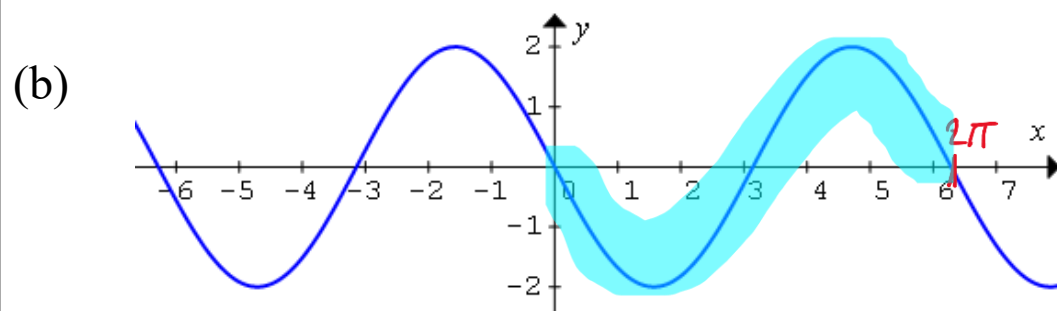
Exercise:

Determine the sine equation for each of the curves:



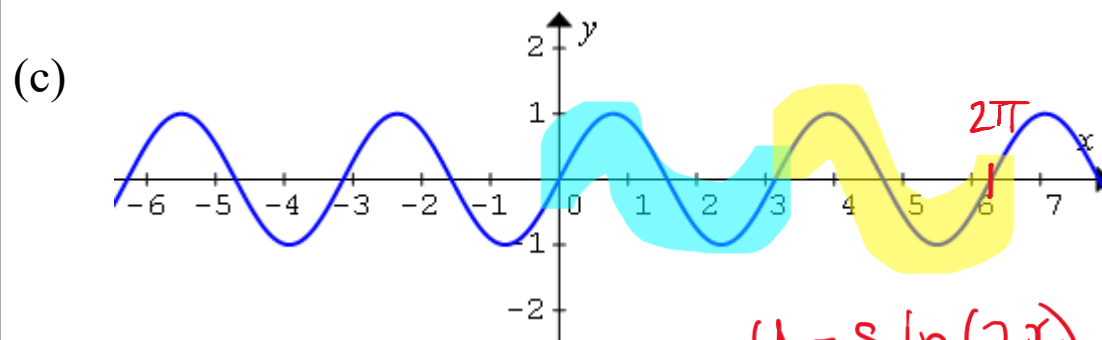
- amplitude = 2
- number of cycles in $2\pi = B = 3$
- no horizontal shift
- no vertical shift

$$y = 2\sin(3x)$$



- amplitude = 2
- number of cycles in $2\pi = B = 1$
- no horizontal shift
- no vertical shift
- graph is flipped upside down

$$y = -2\sin x$$



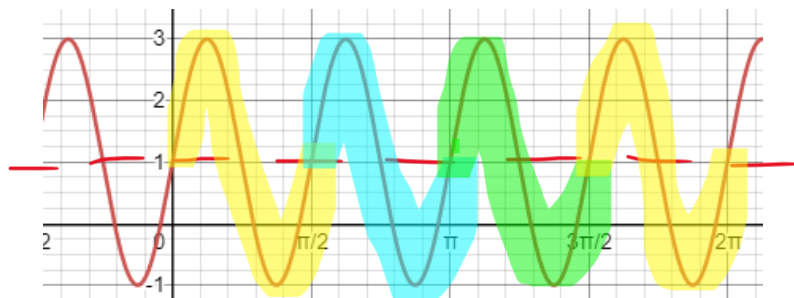
$$y = \sin(2x)$$

- amplitude = 1
- number of cycles in $2\pi = B = 2$
- no horizontal shift
- no vertical shift

Exercise:

For each graph below, determine the amplitude, period and equation.

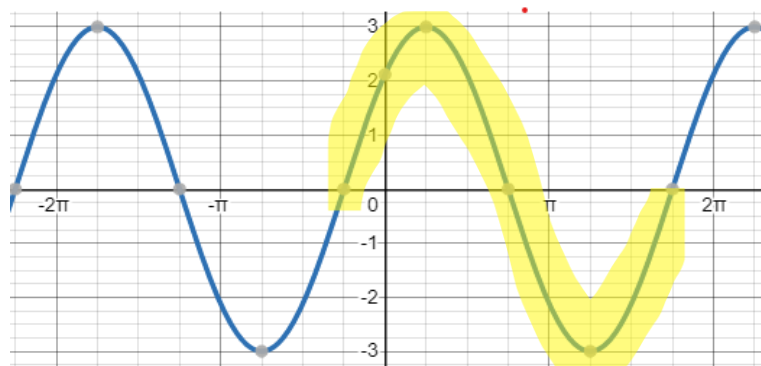
(a)



- amplitude = $A = 2$
- number of cycles in $2\pi = B = 4$
- no horizontal shift
- shifted 1 unit upwards

$$y = 2 \sin(4x) + 1$$

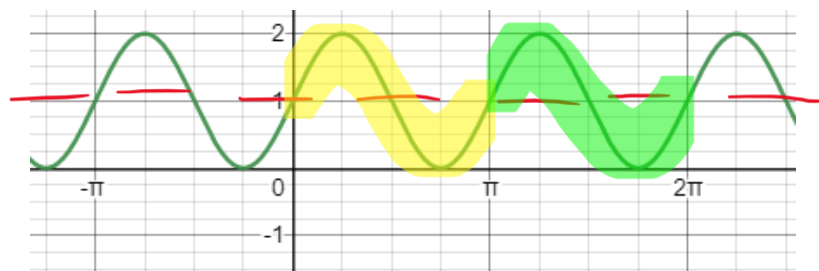
(b)



- amplitude = $A = 3$
- number of cycles in $2\pi = B = 1$
- no vertical shift
- shifted $\frac{\pi}{4}$ to the left

$$y = 3 \sin\left(x + \frac{\pi}{4}\right)$$

(c)



- amplitude = $A = 1$
- number of cycles in $2\pi = B = 2$
- no horizontal shift
- shifted 1 unit upwards

$$y = \sin(2x) + 1$$

Graphs of the Tangent Function

The general form for the tan function is $y = A \tan B(x - h) + k$.

The four constants in the equation (A , B , h and k) will be analysed separately by comparing the graphs of $y = \tan x$ with graphs of $y = A \tan x$, $y = \tan Bx$, $y = \tan(x - h)$ and $y = \tan x + k$.

When $A = 1$, $B = 1$, $h = 0$ and $k = 0$, the equation $y = A \tan B(x - h) + k$ simplifies to $y = \tan x$.

Discussion Questions: <https://www.desmos.com/calculator/61vqbm4ii>

Use Desmos to analyse the transformations in the tan graph by changing values for A , B , h and k .

Use the sliders or the animation feature to study the transformations.

Describe the transformations resulting as each constant is changed:

(d) A (while $B = 1$, $h = 0$, $k = 0$)

makes it steeper & shallower

(b) B (while $A = 1$, $h = 0$, $k = 0$)

spreads it horizontally

(d) h (while $A = 1$, $B = 1$, $k = 0$)

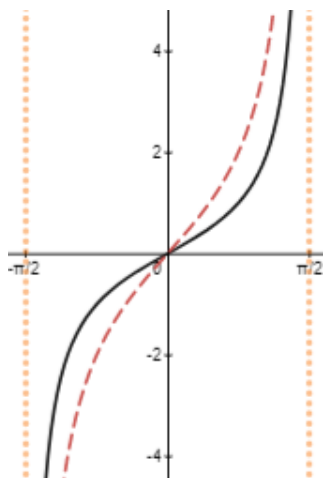
moves it side to side

(d) k (while $A = 1$, $B = 1$, $h = 0$)

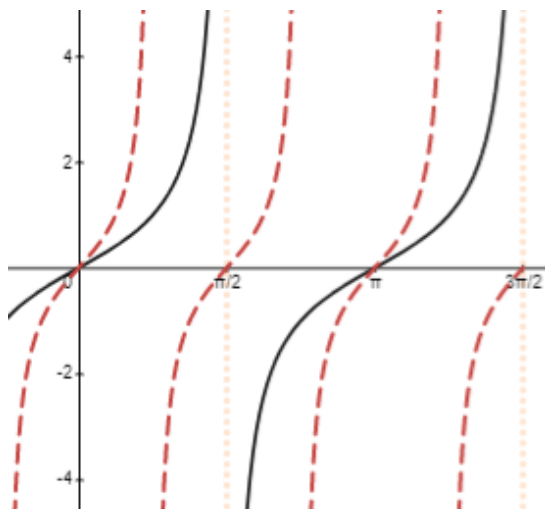
moves it up & down

Examples: As a reference, the parent graph of $y = \tan x$ (solid line) is drawn on each graph.

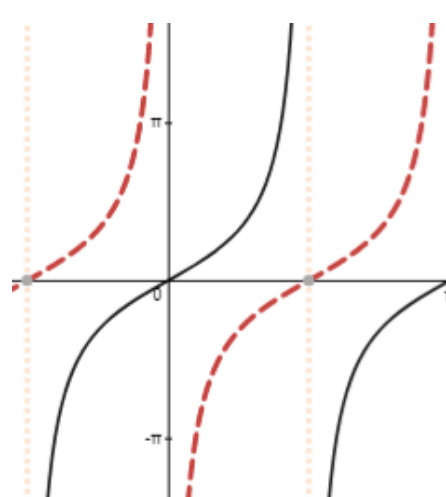
$$y = 2 \tan x$$



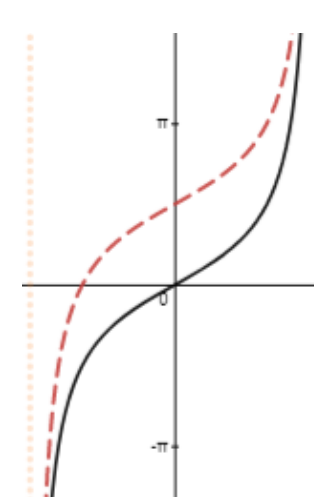
$$y = \tan 2x$$



$$y = \tan\left(x - \frac{\pi}{2}\right)$$



$$y = \tan x + \frac{\pi}{2}$$



$$y = A \tan x$$

The y-values are scaled by a factor of A

$$y = \tan Bx$$

B cycles in interval of π units. The period $p = \frac{\pi}{B}$

$$y = \tan(x - h)$$

h horizontally shifts the graph

$$y = \tan x + k$$

k vertically shifts the graph.